

Total No. of Printed Pages: 2

SUBJECT CODE NO: RNP-8003
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Mechanical) (NEP) Sem - III
Examination May/June-2026
Engineering Thermodynamics

[Time: 3:00 Hours]

[Max. Marks: 60]

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION-A

Q. 1 Solve any Ten

20

- a) PMM-I violates the _____ law of thermodynamics.
- b) Define flow work.
- c) State Kelvin-Planck statement.
- d) Heat engine converts heat into _____.
- e) What is availability?
- f) What is Carnot cycle?
- g) Dead state is when system is in equilibrium with _____.
- h) Carnot cycle has _____ efficiency among all cycles.
- i) Define critical point.
- j) What is dryness fraction?
- k) Steam tables are used to find:
 - a. Pressure only
 - b. Temperature only
 - c. Properties of steam
 - d. Volume only
- l) Orsat apparatus measures:
 - a. Fuel
 - b. Air
 - c. Flue gas
 - d. Steam

SECTION-B

Q. 2

- a) What are the limitations of the first law of thermodynamics? Explain with ex- 3
amples.
- b) A centrifugal pump delivers 3000 kg of water per minute from initial pres- 7
sure of 1 bar to a final pressure of 4.2 bar. The suction is 2.2 m below and
deliver is 8.5 m above the central line of pump. If suction and delivery Pipes
are of 200 mm and 100 mm diameter respectively find the power required to
run the pump.

Q. 3

- a) Prove the equivalence of Kelvin plank & clausius statement. 5
- b) A reversed Carnot cycle operates as refrigerator has capacity of 5

100 kJ/s while operating between temperature limits of -20°C & 35°C Determine power input & COP.

- Q. 4** a) Write short notes on Principle of increase of entropy. **5**
- b) Define availability and irreversibility. Explain how they are related to energy degradation. **5**
- Q. 5** a) Derive the air standard efficiency of the Otto cycle **5**
- b) Write short notes on Atkinson cycle **5**
- Q. 6** a) Explain phase transformation of water at constant pressure. **5**
- b) A pressure cooker contains 2kg of dry & saturated steam at 5bar. Find the quantity of heat that must be rejected so as to reduce quality upto 60% dry. Also find pressure & temp at new state. **5**
- Q. 7** a) Define calorific value. Explain the difference between higher and lower calorific values. **5**
- b) A fuel contains 82% C, 16% H_2 , and 2% O_2 . Calculate the theoretical air-fuel ratio. **5**